



Tecnológico de Monterrey

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EEG processing for NF systems

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Design and Development in Neuroengineering
BI2010B.604
19/04/26

1. EEG Recording

Dataset: DASPS (IEEE DataPort): 23 participants, 6 anxiogenic scenarios each. Subject S01 was used for this analysis. <https://ieee-dataport.org/open-access/dasps-database>

Conditions: Each scenario included two phases: Listen (psychologist narrates a stress scenario aloud) and Recall (participant mentally re-narrates it). The 12 individual 15-second segments were concatenated into a single 3-minute continuous recording.

Relevance to NF challenge: These conditions induce contrasting emotional and cognitive states, making them suitable to evaluate frontal alpha asymmetry as a neuromarker for anxiety monitoring and neurofeedback training.

2. Signal Loaded into EEGLAB

Sampling rate	128 Hz
Channels	14 electrodes (AF3, F7, F3, FC5, T7, P7, O1, O2, P8, T8, FC6, F4, F8, AF4)
Total samples	23,040
Duration	180 s (3 minutes)
Conditions	6 × Listen (odd segments) + 6 × Recall (even segments), 15 s each

3. Band-Pass Filter (1 - 45 Hz): A zero-phase FIR band-pass filter was applied using `pop_eegfiltnew(EEG, 1, 45)` (425-point, transition band ≈ 1 Hz). The lower cutoff at 1 Hz removes DC drift, confirmed by a visible 0 Hz peak in the pre-filter FFT spectrum. The upper cutoff at 45 Hz guards against powerline noise (50/60 Hz), which was not prominent in this dataset, likely pre-attenuated by the headset. The alpha band of interest (8–13 Hz) is fully preserved within the passband.

4. Neuromarker: Frontal Alpha Asymmetry (FAA):

- Rationale: Per Davidson's lateralization model, relatively greater left frontal activation is associated with approach motivation and positive affect, while right-dominant activation indicates avoidance and anxiety. Alpha power is inversely related to cortical activation, so FAA is estimated from alpha power at F3 (left) and F4 (right).
- Formula: $FAA = \ln(\text{mean PSD}_\alpha [F4]) - \ln(\text{mean PSD}_\alpha [F3])$, $\alpha = 8\text{--}13$ Hz
- Computation: Welch's PSD was estimated on each channel using a 2-second sliding window with 75% overlap (0.5 s step). Mean power within 8–13 Hz was extracted per window, log-transformed, and subtracted (right minus left) to produce a continuous FAA trace across the full 180 s recording.
- Results: Mean FAA during Listen = 1.1913; Mean FAA during Recall = 1.3417. Both values are positive (right > left alpha power), indicating relatively greater left frontal activation throughout the session, consistent with an approach-oriented cognitive state. The modestly higher FAA during Recall may reflect increased cognitive engagement during mental imagery of the anxiogenic scenario.

Figure 1 - Raw EEG Signals (F3 & F4)

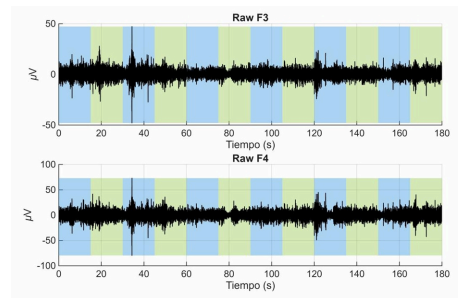


Figure 1. Raw EEG signals at channels F3 (top) and F4 (bottom) across 180 s. Blue regions = Listen; green regions = Recall. Each condition segment is 15 s.

Figure 2 - Filtered EEG Signals (F3 & F4, 1–45 Hz)

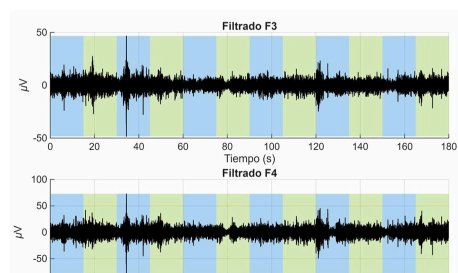


Figure 2. Band-pass filtered (1-45 Hz) EEG signals at F3 (top) and F4 (bottom). Condition shading identical to Figure 1. The change relative to raw is subtle, consistent with minimal spectral content outside the passband beyond DC drift.

Figure 3 - FAA Neuromarker Across Time

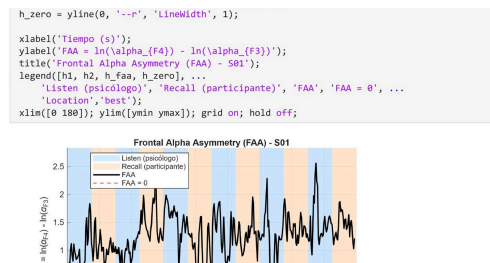


Figure 3. Frontal Alpha Asymmetry ($FAA = \ln(\alpha_{F4}) - \ln(\alpha_{F3})$) estimated with a 2 s sliding window (75% overlap). Blue regions = Listen; orange regions = Recall. Red dashed line at $FAA = 0$. Mean FAA: Listen = 1.1913, Recall = 1.3417.

Conclusion: This analysis successfully implemented a complete EEG processing pipeline for FAA-based neurofeedback targeting emotional state regulation under anxiety. Preprocessing with a 1-45 Hz filter removed DC drift (confirmed via FFT) while preserving the alpha band. FAA was computed continuously from F3/F4 using Welch's method with a 2 s sliding window (75% overlap), yielding a smooth, real-time-capable neuromarker. Both conditions produced positive FAA values, indicating left-dominant frontal alpha throughout; Recall showed a marginally higher mean FAA (1.3417 vs. 1.1913), possibly reflecting the cognitive load of mental imagery rather than a pure anxiety modulation. As the basis for an NF system, this metric provides a physiologically grounded, low-latency signal that could be used to train users toward a target asymmetry profile indicative of reduced anxiety. Future work should include multi-subject validation and source-level analysis to mitigate scalp-level volume conduction artifacts inherent to asymmetry measures.

